
Direct Self-evolving Optimization: Efficient Self-Evolving LLM Training without Challenger Model

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Abstract

1 The self-evolving training paradigm has emerged as a crucial technique for en-
2 hancing the ability of Large Language Models (LLMs) to perform mathematical
3 reasoning and handle complex tasks. By training a challenger model to generate
4 questions for a solver model, this approach effectively overcomes the scarcity of
5 human-annotated data. However, a major drawback of this paradigm is the signifi-
6 cant increase in computational and memory costs. In this paper, we eliminate the
7 need for challenger model training by deriving a closed-form optimal challenger
8 policy, conditioned on the current solver model. We show that this optimal policy
9 is a reweighted version of the base model’s policy, where the weights correspond
10 to the challenger’s reward. Finally, we employ the Metropolis-Hastings algorithm
11 to iteratively sample questions from this optimal policy, which are subsequently
12 fed into the solver model for efficient training.

13 1 Introduction

14 1.1 Existing method

15 In the existing self-evolving LLM training [Huang et al., 2025, Xia et al., 2025], the **challenger**
16 optimizes:

$$\max_{\phi} R_c(\phi, \theta) := \mathbb{E}_{x \sim \pi_{\phi}} [r_c(x, \theta)] - \beta \text{KL}(\pi_{\phi} || \pi_{ref}) \quad (1)$$

17 where $r_c(x, \theta)$ is the challenger reward function given a solver model, e.g., if a generated question x
18 makes solver’s answers more uncertain, challenger will get higher reward.

19 The **solver** optimizes:

$$\max_{\theta} R_s(\phi, \theta) := \mathbb{E}_{x \sim \pi_{\phi}} [\mathbb{E}_{y \sim \pi_{\theta}(y|x)} [r_s(x, y)]] \quad (2)$$

20 where $r_s(x, y)$ is the solver’s reward function.

21 Since the two objectives depend on both ϕ and θ , existing solution is to fix one and update another.
22 Given initialization ϕ_0 and θ_0 , the updating rule is

$$\begin{aligned} \phi_{t+1} &\leftarrow \text{update } \phi_t \text{ for K steps using GRPO on reward } R_c(\phi_t, \theta_t), \\ \theta_{t+1} &\leftarrow \text{update } \theta_t \text{ for K steps using GRPO on reward } R_s(\phi_{t+1}, \theta_t) \end{aligned}$$

23 1.2 The Weakness and Our proposal

24 However, training two models are costly and this alternating optimization is unstable. To optimize
25 equation 1 and equation 2 is to first define

$$\phi^*(\theta) := \arg \max_{\phi} R_c(\phi, \theta),$$

26 Indeed the closed form solution is

$$\pi_{\phi^*}(\theta) = \frac{1}{Z(\theta)} \pi_{ref}(x) \exp\left(\frac{r_c(x, \theta)}{\beta}\right)$$

27 then plug $\pi_{\phi^*}(\theta)$ into equation 2:

$$\begin{aligned} \max_{\theta} R(\theta) &:= R_s(\phi^*(\theta), \theta) \\ &= \mathbb{E}_{x \sim \pi_{\phi^*}} [\mathbb{E}_{y \sim \pi_{\theta}} [r_s(x, y)]] = \sum_x \pi_{\phi^*}(\theta)(x|p_0) \mathbb{E}_{x \sim \pi_{\theta}} [r_s(x, y)] \\ &= \sum_x \frac{1}{Z(\theta)} \pi_{ref}(x|p_0) \exp\left(\frac{r_c(x, \theta)}{\beta}\right) \mathbb{E}_{y \sim \pi_{\theta}(y|x)} [r_s(x, y)] \\ &= \frac{1}{Z(\theta)} \mathbb{E}_{x \sim \pi_{ref}(x|p_0)} \left[\exp\left(\frac{r_c(x, \theta)}{\beta}\right) \mathbb{E}_{y \sim \pi_{\theta}(y|x)} [r_s(x, y)] \right] \end{aligned} \quad (3)$$

28 where $r_c(x, \theta) = \max\{0, r_{\text{uncertainty}}(x; \theta) - r_{\text{rep}}(x)\}$, $r_{\text{uncertainty}}(x; \theta) := 1 - 2|\hat{p}(x; S_{\theta}(x)) - \frac{1}{2}|$,
29 and $\hat{p}(x; S_{\theta}(x)) = \frac{1}{m} \sum_{j=1}^m \mathbb{I}\{y_j = \bar{y}(x)\}$, and $S_{\theta}(x) = \{y_1, \dots, y_m\}$ is the set of m responses that
30 solver model θ generated for question x , and $\bar{y}(x)$ is the most frequent answer in $S_{\theta}(x)$.

31 $r_{\text{rep}}(x)$ is the repetition penalty. Given a batch of questions $x_1, \dots, x_B$, we compute pairwise
32 distances using BLEU score similarity, $d_{ij} = 1 - BLEU(x_i, x_j)$, and group questions where
33 $d_{ij} < \tau_{BLEU}$ into clusters $C = \{C_1, \dots, C_K\}$. The penalty for a question x_i in a cluster C_k is
34 proportional to its relative size:

$$r_{\text{rep}}(x) = \frac{|C_k|}{B}.$$

35 For $r_s(x, y_j)$, it is 1 if y_j is identical to the pseudo label $\bar{y}(x)$, else 0.

36 Hence, we can directly optimize this objective using policy gradient. First, we generate a mini-batch
37 of questions from the base model $\pi_{ref}: \{x_1, \dots, x_G\}$. For each question, the solver model generates
38 m responses: $A = \{y_{i,1}, \dots, y_{i,m}\}$. Then we estimate the policy gradient as

$$\hat{\nabla}_{\theta} R(\theta) = \frac{1}{G} \sum_{i=1}^G \exp\left(\frac{r_c(x_i, \theta)}{\beta}\right) \frac{1}{m} \sum_{j=1}^m \nabla_{\theta} \log \pi_{\theta}(y_{i,j}|x_i) \quad (4)$$

39 We directly optimize the solver model. It is a weighted policy gradient method, where the weights
40 depend on how the current solver model performs on the generated questions. For the questions that
41 the solver has higher uncertainty, its gradient will be up-weighted.

42 1.3 Active Sampling

43 The objective in equation 3 can be equivalently written as:

$$\max_{\theta} R(\theta) = \mathbb{E}_{x \sim \pi_{\phi^*}(\theta)(x|p_0)} [\mathbb{E}_{y \sim \pi_{\theta}(y|x)} [r_s(x, y)]]$$

44 where $\pi_{\phi^*}(\theta) \sim \pi_{ref}(x) \exp\left(\frac{r_c(x, \theta)}{\beta}\right)$. Now the question is, how can we actively sample questions
45 from $\pi_{\phi^*}(\theta)(x|p_0)$ only given access to π_{base} ? A generally used approach is MCMC sampling via
46 Metropolis-Hasting algorithm.

Algorithm 1: MCMC-DEO

Input: θ_0 , question-generating prompt p_0 , learning rate η , regularization parameter β .

for $t = 0, \dots, T - 1$ **do**

 Sample N questions $\{x_1^0, \dots, x_N^0\}$ from $\pi_{base}(\cdot|p_0)$.

 For each x_i^0 , generate mutated version $\hat{x}_i = MCMC(x_i^0, \tau, \pi_{base})$ For each $\hat{x}_i$, generate m answers $\{y_{i,1}, \dots, y_{i,m}\}$ from $\pi_{\theta_t}(\cdot|x_i)$.

 Compute mini-batch gradient $\hat{\nabla}_{\theta} R(\theta) = \frac{1}{N} \sum_{i=1}^N \frac{1}{m} \sum_{j=1}^m \nabla_{\theta} r_s(x_i, y_{i,j}) \log \pi_{\theta}(y_{i,j}|x_i)$.

$\theta_{t+1} = \theta_t + \eta \hat{\nabla}_{\theta} R(\theta)$.

end

Output: θ_T

Algorithm 2: MCMC(x_0, τ, π)

Input: Original question x_0 , mutation prompt p , policy π

for $t = 0, \dots, \tau - 1$ **do**

 Sample a mutated question $x_{t+1} \sim \pi(x_t|p)$,

 Compute acceptance probability $\alpha \doteq \min\{1, \exp(r_c(x') - r_c(x))\}$

 Accepting $x_{t+1} = x'$ with probability α , otherwise $x_{t+1} = x_t$.

end

Output: x_{τ}

47 1.4 Adaptive Temperature Update

48 We wish to control the hardness of the proposed questions by restricting their uncertainty
49 score r_{unc} within the range $[r_{\min}, r_{\max}]$. Define δ to be the violation rate. $r_c(X; \theta) =$
50 $\max\{0, \sum_{x \in X} r_{\text{uncertainty}}(x; \theta) - \lambda r_{\text{rep}}(X)\}$. This can be cast as a constrained optimization problem
51 for β :

$$\begin{aligned} & \max_{\beta \in [\beta_{\min}, \beta_{\max}]} \beta \\ & \text{subject to: } 1 - \mathbb{E}_{X \sim q(\beta)}(V(X)) \leq \delta \end{aligned} \quad (5)$$

52 where $V(X) := \frac{1}{N} \sum_{x \in X} \mathbf{1}\{r_{\min} \leq r_{unc}(x; \theta) \leq r_{\max}\}$.

53 It is equivalent to solving:

$$\begin{aligned} & \min_{\beta \in [\beta_{\min}, \beta_{\max}]} -\log \beta \\ & \text{subject to: } 1 - \mathbb{E}_{X \sim q(\beta)}(V(X)) \leq \delta. \end{aligned} \quad (6)$$

54 The Lagrange minimax problem is:

$$\min_{\beta \in [\beta_{\min}, \beta_{\max}]} \max_{\lambda \geq 0} \mathcal{L}(\beta, \lambda) = -\log \beta + \lambda(1 - \mathbb{E}_{X \sim q(\beta)}(V(X)) - \delta)$$

55

$$\nabla_{\beta} \mathcal{L} = -\frac{1}{\beta} - \lambda \frac{d}{d\beta} \mathbb{E}_{X \sim q(\beta)}(V(X)) \quad (7)$$

$$= -\frac{1}{\beta} - \lambda \frac{d}{d\beta} \sum_{X \in \mathcal{X}} q_{\beta}(X)(V(X)) \quad (8)$$

$$(9)$$

56 since $\log q_{\beta}(X) = \log \pi_{base}(X) + \frac{r_c(X, \theta)}{\beta} - \log Z(\theta, \beta)$

57 So

$$\frac{\partial}{\partial \beta} \log q_{\beta}(X) = -\frac{r_c(X, \theta)}{\beta^2} + \frac{1}{\beta^2} \mathbb{E}_{q(\beta)}[r_c(X, \theta)]$$

58 Hence

$$\frac{\partial q_{\beta}(X)}{\partial \beta} = q_{\beta}(X) \frac{\partial}{\partial \beta} \log q_{\beta}(X) = -\frac{q_{\beta}(X)}{\beta^2} (r_c(X, \theta) - \mathbb{E}_{q(\beta)}[r_c(X, \theta)])$$

59 Hence

$$\nabla_{\beta} \mathcal{L} = -\frac{1}{\beta} - \lambda \sum_{X \in \mathcal{X}} -\frac{q_{\beta}(X)}{\beta^2} (r_c(X, \theta) - \mathbb{E}_{q(\beta)}[r_c(X, \theta)]) (V(X)) \quad (10)$$

$$= -\frac{1}{\beta} + \frac{\lambda}{\beta^2} (\mathbb{E}_{q(\beta)}[r_c(X, \theta)V(X)] - \mathbb{E}_{q(\beta)}[r_c(X, \theta)]\mathbb{E}_{q(\beta)}[V(X)]) \quad (11)$$

60 This gradient can be estimated by B mcmc samples: $X_1, \dots, X_B$, and compute $\bar{V} = \frac{1}{B} \sum_{i=1}^B V(X_i)$,
 61 $\bar{R} = \frac{1}{B} \sum_{i=1}^B r_c(X_i, \theta)$:

$$\hat{\nabla}_{\beta} \mathcal{L} = -\frac{1}{\beta} + \frac{\lambda}{\beta^2} \frac{1}{B-1} \sum_{i=1}^B (V(X_i) - \bar{V})(R(X_i) - \bar{R}).$$

62 The gradient for λ is

$$\nabla_{\lambda} \mathcal{L} = (1 - \mathbb{E}_{X \sim q(\beta)}(V(X)) - \delta)$$

63 the empirical version is

$$\hat{\nabla}_{\lambda} \mathcal{L} = 1 - \frac{1}{B} \sum_{i=1}^B V(X_i) - \delta$$

64 Then, running few steps of gradient descent ascent to update β :

$$\begin{aligned} \beta^{t+1} &= \mathcal{P}_{[\beta_{\min}, \beta_{\max}]}(\beta^t - \eta_{\beta} \hat{\nabla}_{\beta} \mathcal{L}(\beta^t, \lambda_t)) \\ \lambda^{t+1} &= [\lambda^t + \eta_{\lambda} \hat{\nabla}_{\lambda} \mathcal{L}(\beta^t, \lambda_t)]_+ \end{aligned}$$

65 References

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Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

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